$$|\Psi(t)\rangle = \sum_j \alpha_j(t)|E_j\rangle.$$

7. Using Eq. 4.30, replace each $\alpha_j(t)$ with $\alpha_j(0)e^{-\frac{i}{\hbar}E_j t}$:

$$|\Psi(t)\rangle = \sum_j \alpha_j(0)e^{-\frac{i}{\hbar}E_j t}|E_j\rangle. \qquad (4.34)$$

8. Season according to taste.

We can now predict the probabilities for each possible outcome of an experiment as a function of time, and we are not restricted to energy measurements. Suppose **L** has eigenvalues λ_j and eigenvectors $|\lambda_j>$. The probability for outcome λ is

$$P_\lambda(t) = |\langle\lambda|\Psi(t)\rangle|^2.$$

Exercise 4.6: Carry out the Schrödinger Ket recipe for a single spin. The Hamiltonian is $\mathbf{H} = \frac{\omega\hbar}{2}\sigma_z\sigma_z$ and the final observable is σ_x. The initial state is given as $|u>$ (the state in which $\sigma_z = +1$).

After time t, an experiment is done to measure σ_y. What are the possible outcomes and what are the probabilities for those outcomes?

Congratulations! You have now solved a real quantum mechanics problem for an experiment that can actually be carried out in the laboratory. Feel free to pat yourself on the back.